# Network for Aging Research Catalyst Grant Application Package Spring 2025

***Please complete all pages in full and submit to Liz Drolle (nar@uwaterloo.ca)***

***by Monday, April 18, 2025 by 4:00PM***

# Applicant Details

| **Principal Applicant (Contact Person)** | |
|---|---|
| Full name: Troy D. Glover | |
| Department: Recreation & Leisure Studies | Faculty: Health |
| Email: troy.glover@uwaterloo.ca | UW Extension: 43097 |
| Specify the nature of your appointment (tenured or tenure-track, definite term): Tenured | |
| **Principal Applicant (if project is co-led by PIs from two Faculties)** | |
| Full name: Jennifer Dean | |
| Department: School of Planning | Faculty: Environment |
| Email: jennifer.dean@uwaterloo.ca | UW Extension: 49107 |
| Specify the nature of your appointment (tenured or tenure-track, definite term): Tenured | |
| **Co-Principal Applicant** ☐ | **Co-Applicant** ☐ |
| Full name: Steve Fischer | |
| Department: Kinesiology & Health Sciences | Faculty: Health |
| Email: steven.fischer@uwaterloo.ca | UW Extension: 42368 |
| Please specify if the co-applicant is a faculty member, PhD student or postdoctoral fellow: Faculty member | |
| **Co-Principal Applicant** ☐ | **Co-Applicant** ☐ |
| Full name: Leia Minaker | |
| Department: School of Planning | Faculty: Environment |
| Email: lminaker@uwaterloo.ca | UW Extension: 45615 |
| Please specify if the co-applicant is a faculty member, PhD student, postdoctoral fellow: Faculty member | |
| | |
| | |

| | | |
|---|---|---|
| **Signatures:** Provide the name and signature of both the applicant(s) and the co-applicant(s) who have confirmed their willingness to play a significant role in the proposed project (scanned or electronic)<br><br>NOTE: If you are successful in receiving a NAR grant, your information (name, contact and project information) will be posted on the NAR website and used in promotions from NAR. | | |
| **Principal Applicant name:**<br>Troy Glover | **Signature:** | **Date:** |
| | | |
| **Principal applicant name:**<br>Jennifer Dean | **Signature:** | **Date:**<br>April 28, 2025 |
| | | |
| **Co-applicant name:**<br>Steve Fischer | **Signature:**<br>Steven Fischer Digitally signed by Steven Fischer Date: 2025.04.28 13:49:24 -04'00' | **Date:**<br>April 28, 2025 |
| | | |
| **Co-applicant name:**<br>Leia Minaker | **Signature:**<br>Leia Minaker Digitally signed by Leia Minaker Date: 2025.04.28 13:57:15 -04'00' | **Date:**<br>April 28 2025 |

# Project Description

Project Title:

Enhancing Social Connectedness Among Older Adults in the START-FIT Program

| Start date: | End Date: |
| --- | --- |
| July 1, 2025 | June 30, 2026 |

Is this new or exploratory research? Or are you examining a new aspect of a current or previous project?

This project is new and exploratory.

Have you received funding from the NAR Seed Grant program previously? If yes, when?

No

*Preference will be given to researchers who have not previously received funding from this program. If you have received funding previously, please attach a one page summary of any outcomes from the project that was funded. This should include activities completed and any publications or subsequent grants that resulted from the project.*

*Researchers who have been previously funded by NAR but have not submitted their final report within the specified timelines are not eligible for further support from NAR.*

**Background and Justification**

*Be sure to demonstrate a strong relevance to, and convincing rationale for, the importance of your research and why it is a priority specifically related to aging. You should clearly expand on the impact your research is expected to have in the field of aging.*

Social isolation and loneliness among older adults represent a growing public health crisis with well-documented consequences, including increased risk of chronic disease, cognitive decline, depression, and premature death (Holt-Lunstad, 2022; WHO, 2023). These risks are especially pronounced for older adults undergoing cancer treatment, who often face physical limitations, psychosocial stressors, and disruptions to their social routines (Zhao at al., 2024). In this context, innovative interventions that promote social connection are urgently needed to improve health outcomes and support aging in place (Canadian Alliance for Social Connection and Health, 2024; O'Rourke et al., 2018).

Cancer treatment often leads to physical fatigue, social withdrawal, and emotional distress—conditions exacerbated in later life that can contribute to increased isolation and loneliness (Raque-Bogdan et al., 2019). While exercise is a well-established intervention that mitigates the physical effects of cancer treatment (Bell et al., 2021; Noble et al., 2012), its potential role in rebuilding social ties and preventing social isolation has not been adequately explored within clinical rehabilitation contexts (see Shvedko et al., 2018). This represents a significant research gap, especially given that older adults' ability to age in place is closely tied to their social connectedness and the quality of their built environment.

The START-FIT program—a supervised 12-week in-treatment cancer exercise intervention delivered at CCCARE—provides an ideal setting to pilot a social connectedness intervention. With its structured, twice-weekly sessions, consistent staffing, and rolling admission of small cohorts, the program already fosters conditions conducive to community-building. However, currently, there are no group-based social components nor any formal assessment of social outcomes. This project seeks to leverage and expand the START-FIT model to explore how deliberate social interventions, combined with attention to built environment supports, can improve both short- and long-term outcomes for older adults.

This project is particularly aligned with priorities in aging research that seek to promote aging in place, reduce health system burdens, and enhance community resilience. It responds to a call for scalable, cost-effective, and socially integrated models of care for older adults with complex health needs. By focusing on social infrastructure and built environment factors that support continued engagement, post-program, in health-promoting activities, the findings will inform local planning, program design, and policy discussions about healthy aging.

This proposed research has the potential to reshape how we think about rehabilitation programs for older adults—not just as spaces for physical healing, but as vital hubs for rebuilding social connection and community engagement. By embedding social interventions into exercise programs and linking them with planning and environmental data, we can design better, more sustainable health promotion strategies that support aging in place and community belonging.

**Proposed Research**

*Provide detailed description and justification for your research objectives and methodology with an emphasis on scientific excellence. Make a case for the novel or exploratory nature of the research you are proposing, as well as its feasibility.*

OBJECTIVES: The purpose of the proposed research is to investigate how the integration of social connectivity elements into the START-FIT program, a 12-week supervised exercise program that targets adults undergoing active cancer treatments, can enhance and sustain aging-in-place for older adults living with cancer. Accordingly, the objectives of the proposed research are to: (1) design and pilot a social connectedness intervention within START-FIT, incorporating group warm-ups, paired walking activities, and facilitated social check-ins; (2) evaluate changes in participants' social connectedness, loneliness, and sustained exercise adherence following the intervention; (3) examine the influence of built environment factors (e.g., proximity to parks, community centres, walkability) on participants' post-program physical and social engagement; and (4) identify planning and policy strategies that enhance social and physical activity infrastructure to support aging in place.

METHODOLOGY: This study will use a mixed-methods, quasi-experimental design to evaluate the effect of a social connectedness intervention embedded within START-FIT. Two cohorts of older adults (65+), each with 10 participants, will be recruited from the START-FIT program. One cohort will receive standard programming, while the other will engage in added social components co-designed with CCCARE staff. Quantitative data will be gathered pre- and post-program using validated tools, namely the UCLA 3-Item Loneliness Scale (Hughes et al., 2004), Friendship Scale (Hawthorne, 2006), and PAGEQ (Estabrooks & Carron, 2000), along with attendance and adherence logs. To explore the evolution and structure of social relationships, participants will also complete egocentric network maps (see Altissimo, 2016) at baseline and after 12 weeks. These maps will serve as prompts during post-program interviews to help participants reflect on changes in their social networks, perceived support, and opportunities for continued connection and activity. To complement this approach, spatial analysis will use GIS to map participants' proximity to parks, trails, and transit, helping us assess environmental supports or constraints for sustaining activity and connection. Qualitative interview data and network maps will be analyzed thematically to identify relationship patterns, while quantitative results will be explored using descriptive and inferential statistics. This integrated approach allows us to assess both the social and spatial dynamics influencing older adults' ability to remain active and engaged following cancer treatment.

NOVELTY & FEASIBILITY: This project is among the first to apply a qualitative social network analysis within a clinical cancer exercise setting, offering an innovative way to understand the social benefits of group-based rehabilitation. By combining participant-generated network maps with interview narratives and GIS mapping, the study provides a comprehensive view of how social and environmental factors intersect. START-FIT's small-group structure and rolling intake make it an ideal platform for piloting a social intervention with immediate relevance to health and planning professionals. CCCARE's strong interest in evaluating social outcomes and its capacity for program adaption further enhance the feasibility of this timely, low-cost initiative.

**The Research Team**

*Explain how the expertise of this team will be complementary in completing this research.*

The research team brings together a dynamic blend of disciplinary expertise spanning recreation, health, planning, and clinical rehabilitation. Dr. Troy Glover, Professor in the Department of Recreation and Leisure Studies, serves as the Principal Applicant and contributes extensive knowledge in social capital, placemaking, and social infrastructure, with a particular focus on community-based approaches to reducing isolation and promoting connection. Dr. Jennifer Dean, Associate Professor in the School of Planning, offers research strengths in healthy community design and aging in place, supporting both the conceptual framing and the spatial analysis elements of the study. Dr. Steven Fischer, Director of CCCARE and a leading expert in clinical exercise science, provides deep expertise in cancer rehabilitation and oversees the START-FIT program where the study will be embedded. Dr. Leia Minaker, also an Associate Professor in Planning, brings applied public health and policy experience, with specific attention to the built environment and systems-level strategies that influence community health. Together, the team's expertise is highly complementary. This combination of skills and perspectives ensures the methodological, theoretical, and practical rigor required to execute a successful pilot project with strong potential for future scalability.

**HQP Training Plan**

*Details are required to show how HQP trainees will be directly involved in a meaningful way. Contributions may include such activities as data collection and analysis, report writing, etc., but would exclude tasks that are focused solely on administration and/or coordination such as data entry, meeting scheduling, minute taking, distribution of paperwork*

This project will provide an exceptional training opportunity for one graduate student at the University of Waterloo. The student will be hired as an RA and will engage in multiple aspects of the research process, offering a rich, hands-on learning experience consistent with the Tri-Council's emphasis on developing highly qualified personnel (HQP) through applied, interdisciplinary research. Under the guidance of the project's faculty investigators, the HQP will be responsible for administering surveys and collecting and entering data. They will also participate in the transcription and thematic coding of qualitative interview data, helping to explore how social connections form and evolve within the START-FIT program. In addition, the student will contribute to qualitative network mapping analysis and help integrate these findings with insights from the GIS-based spatial analysis. This blend of qualitative and spatial methods will expose the student to innovative and transferable research techniques that are increasingly valued in health, planning, and community research. The HQP will also support the co-development of knowledge translation materials and will be invited to co-author an academic manuscript for submission to a peer-reviewed journal. In doing so, the student will gain skills in academic writing, public communication, and collaborative interdisciplinary work, skills essential to building research leadership capacity in Canada.

**Plans for External Funding**

*Explain the expected outcomes in terms of future contributions to an external grant application.*

This Catalyst Grant will provide critical pilot data to support future external funding applications aimed at scaling and deepening the proposed intervention. The interdisciplinary nature of the project—integrating health, planning, and social science—makes it highly suitable for tri-council funding opportunities. Specifically, we intend to use the results from this study to develop a full proposal for a CIHR Project Grant, potentially under the Healthy Cities Research Initiative, which supports cross-sector collaborations to promote healthy aging and community resilience. The study's attention to the built environment and social infrastructure also aligns well with SSHRC's Insight Grant and Partnership Development Grant streams, both of which prioritize research on inclusive, sustainable, and connected communities. By demonstrating proof-of-concept through this Catalyst Grant, we aim to justify a larger-scale implementation and evaluation of the social connectedness intervention across multiple community and clinical contexts. We will also seek funding through provincial health research bodies and municipal partnerships to explore policy applications and long-term sustainability. The pilot data will strengthen the competitiveness of these future applications by offering concrete evidence of feasibility, interdisciplinary collaboration, and early-stage impact.

**Knowledge Translation Plan**

*Demonstrate that knowledge users are involved in a meaningful and integrated manner. A knowledge translation plan should be provided; this may include, but is not limited to, communication of research findings to the scientific community or lay audiences, or engagement of relevant knowledge users at various points throughout the research project. Knowledge users may include older adults, families, decision-makers, industry representatives, health care practitioners, or other non-researcher stakeholders who may inform the study process or apply the results in practical settings.*

This project will follow an integrated knowledge translation (KT) approach, engaging key knowledge users—including CCCARE staff, START-FIT participants, municipal planners, and policymakers—throughout the research process. From the co-design of the intervention to the interpretation and dissemination of findings, these stakeholders will help shape the direction and utility of the project. Findings will be shared through multiple formats tailored to different audiences to ensure accessibility and real-world impact. We will develop a visually engaging infographic and a brief summary report that highlights key outcomes and recommendations for both practitioners and participants. The research team will present findings at community meetings and to municipal and regional planning bodies to encourage dialogue around aging, built environments, and social infrastructure. Academic dissemination will include a co-authored manuscript for submission to a peer-reviewed journal such as Supportive Care in Cancer or Journal of Aging and Health, as well as presentations at relevant conferences, including the Canadian Association on Gerontology and the Ontario Public Health Association. We also plan to use institutional and local media channels to raise broader public awareness of the project's goals and contributions. This KT strategy ensures that the knowledge generated through this study informs not only academic debates, but also local planning, service delivery, and policy initiatives that support healthy aging.

**Proposal Budget and Justification**

*Eligible expenses include travel, meeting costs, personnel, equipment, operating costs, and research-related expenses.*

Total Funds Requested: $10,000

Personnel Costs: $7,832

The majority of the funds requested will be used to support one PhD student from the Department of Recreation and Leisure Studies through a 0.5 Research Assistantship during the Fall 2025 academic term. This assistantship will follow the University of Waterloo's recommended salary and benefits rate for doctoral students. A PhD student is well suited for this role due to the complex, interdisciplinary nature of the proposed project, which includes both qualitative and spatial analysis. The student will contribute to all aspects of the research process, including participant recruitment, survey administration, facilitation and transcription of interviews, thematic analysis, qualitative social network mapping, and GIS-informed spatial analysis. The student will also play an active role in the development of knowledge translation materials and be encouraged to co-author any academic manuscripts that result from the project.

Materials and Supplies: $600

Funds are allocated for the printing of consent forms, survey instruments, and recruitment materials, as well as postage and materials needed to support communication with participants.

Software and Data Access: $1,000

This amount will be used to support access to GIS and spatial analysis tools, including any required licenses and data purchases. These tools are critical for mapping participants' proximity to social infrastructure and understanding the built environment's role in supporting physical activity and connection.

Knowledge Translation: $568

Funds are allocated for the production of knowledge translation materials, including infographics, summary reports, and printed briefs tailored for community partners, municipal planners, and academic audiences. These materials will help ensure that the findings are disseminated beyond academic circles and inform practice and policy.

In-Kind Contributions: In-kind contributions from the research team include CCCARE's program space and personnel support for implementing the intervention, faculty mentorship across disciplines, and access to spatial data analysis tools through the School of Planning. These contributions strengthen the feasibility and interdisciplinary value of the project.

**References**

Bell, K. E., Pfeiffer, A. G., Schmidt, S., Bos, L., Russell, C., Barnes, T., ... & Mourtzakis, M. (2021). Low-frequency exercise training improves cardiovascular fitness and strength during treatment for breast cancer: A single-arm intervention study. Scientific Reports, 11 (1), 22758.

Canadian Alliance for Social Connection and Health (2024). Public Health Guidelines for Social Connection. Available from: https://www.socialconnectionguidelines.org/en/

Estabrooks, P. A., & Carron, A. V. (2000). The Physical Activity Group Environment Questionnaire: An instrument for the assessment of cohesion in exercise classes.Group Dynamics: Theory, Research, and Practice, 4(3), 230–243.

Hawthorne, G. (2006). Measuring social isolation in older adults: development and initial validation of the friendship scale. Social indicators research, 77, 521-548.

Holt-Lunstad, J. (2022). Social connection as a public health issue: the evidence and a systemic framework for prioritizing the “social” in social determinants of health. Annual Review of Public Health, 43(1), 193-213.

Hughes, M. E., Waite, L. J., Hawkley, L. C., & Cacioppo, J. T. (2004). A short scale for measuring loneliness in large surveys: Results from two population-based studies. Research on aging, 26(6), 655-672.

Noble, M., Russell, C., Kraemer, L., & Sharratt, M. (2012). UW WELL-FIT: the impact of supervised exercise programs on physical capacity and quality of life in individuals receiving treatment for cancer. Supportive Care in Cancer, 20, 865-873.

O’Rourke, H. M., Collins, L., & Sidani, S. (2018). Interventions to address social connectedness and loneliness for older adults: a scoping review. BMC geriatrics, 18, 1-13.

Raque-Bogdan, T. L., Lamphere, B., Kostiuk, M., Gissen, M., & Beranek, M. (2019). Unpacking the layers: a meta-ethnography of cancer survivors’ loneliness. Journal of Cancer Survivorship, 13, 21-33.

Shvedko, A., Whittaker, A. C., Thompson, J. L., & Greig, C. A. (2018). Physical activity interventions for treatment of social isolation, loneliness or low social support in older adults: A systematic review and meta-analysis of randomised controlled trials. Psychology of Sport and Exercise, 34, 128-137.

WHO. (2023). WHO launches commission to foster social connection. Available from: https://www.who.int/news/item/15-11-2023-who-launches-commission-to-foster-social-connection

Zhao, J., Reese, J. B., Han, X., & Yabroff, K. R. (2024). Loneliness and mortality risk among cancer survivors in the United States: a retrospective, longitudinal study. Journal of the National Comprehensive Cancer Network, 22(4), 244-248.